

Hands-on 1: Introduction to HQL and JPQL (Theory)

1. What is JPQL?

JPQL (Java Persistence Query Language) is a query language used in JPA (Java Persistence API) to perform database operations using **entity objects instead of database tables**.

It is designed to be database-independent and works on entity models.

Key Features of JPQL:

- Works with entity class names
- Works with entity fields (not table columns)
- Platform independent
- Supports SELECT, UPDATE, DELETE operations
- Does NOT support INSERT directly

2. What is HQL?

HQL (Hibernate Query Language) is an object-oriented query language used in Hibernate framework.

It is similar to JPQL but is more powerful and Hibernate-specific.

Key Features of HQL:

- Works with Java entity classes
- Database independent
- Supports SELECT, UPDATE, DELETE
- Additionally supports INSERT statements
- Provides advanced features like:
 - Fetch joins
 - Polymorphic queries
 - Bulk operations

3. Relationship between HQL and JPQL

Feature	JPQL	HQL
Defined by	JPA specification	Hibernate framework
Scope	Subset of HQL	Superset of JPQL

Feature	JPQL	HQL
Portability	High	Hibernate-specific
INSERT support	Not supported	Supported
SELECT/UPDATE/DELETE	Supported	Supported

4. Important Concept

JPQL is a subset of HQL

This means:

All JPQL queries will work in HQL

But all HQL queries may not work in JPQL

5. Similarity with SQL

Both HQL and JPQL:

- Look similar to SQL
- Use SELECT, FROM, WHERE, JOIN

But difference is:

SQL	JPQL / HQL
Works with tables	Works with entities
Works with columns	Works with fields
Database dependent	Database independent

6. Example Comparison

SQL Query

```
SELECT * FROM employee WHERE salary > 50000;
```

JPQL / HQL Query

```
SELECT e FROM Employee e WHERE e.salary > 50000
```

Notice:

- Employee is entity name (not table)
- salary is Java field (not column)

7. Operations supported

Both HQL and JPQL support:

- SELECT
- UPDATE
- DELETE

Only HQL supports:

- INSERT (bulk insert operations)

Example:

```
INSERT INTO Employee(name, salary)
SELECT e.name, e.salary FROM Employee e
```

8. Why HQL / JPQL is used?

- ✓ Avoids writing database-specific SQL
- ✓ Works across multiple databases
- ✓ Uses object-oriented approach
- ✓ Easier integration with Hibernate/JPA
- ✓ Reduces dependency on table structure

9. Summary

- JPQL is a standard query language of JPA
- HQL is Hibernate's extended version of JPQL
- Both are object-oriented query languages
- They use entity names instead of table names
- JPQL is portable, HQL is more powerful

JPQL and HQL are object-oriented query languages used to perform database operations using entity classes instead of tables, where JPQL is a subset of HQL and HQL provides additional Hibernate-specific features